

Report on Glacier Data Prediction

TENG ZHANG

1. INTRODUCTION

This report presents the data processing and modeling workflow applied to the glacier grid dataset, which includes:

- Raw data ingestion and cleaning
- Time-based split: training set / test set
- EDA
- DATA PREPROCESSING
- Model training and comparison: Linear Regression, Lasso, SVR, Decision Tree, Random Forest, MLP
- Using Genetic Algorithm (GA) to Optimize Feature Selection

The code and results are available on GitHub (<https://github.com/ten-z/Evolutionary-Machine-Learning>). Please note that the High-res-more-variables dataset was too large to process locally within the available time, so it has not been included in this report.

2. RAW DATA INGESTION AND CLEANING

2.1. Sequential Reading and Initial Cleaning

Read each year's raw gridded data in chronological order.

Remove any rows containing the fill value (9.969...e+36) or NaN, which correspond almost entirely to border cells over open ocean.

2.2. Local Neighborhood Feature Construction

Manually specify one of ice_thickness, ice_velocity, or ice_mask as the prediction target ("label").

For each remaining grid cell, collect all non-coordinate variables from its eight surrounding neighbors.

If a neighbor has been removed (open ocean), substitute zeros for its features.

Append these neighborhood values as new features to the central cell.

2.3. Historical Feature Augmentation

For each grid cell, retrieve the same set of variables at the identical location over the past K years (default `past_years = 3`).

If a previous year's data is unavailable, fill those historic features with zeros.

Append these K -year historical values as additional features.

2.4. Year Feature and Final Assembly

Combine the processed data from all years into a single table.

Insert a new year column at the front of every row to record its original year.

Export the consolidated dataset to `{label}_all_years.csv`.

3. TIME-BASED SPLIT: TRAINING SET / TEST SET

Sort the data in ascending order by year, then split it into a training set and a test set according to the `TRAIN_RATIO` (default 0.8), ensuring that all training-set years come before any test-set years and that at least one year is reserved for testing. Finally, save the training and test sets to separate CSV files.

4. EDA

4.1. Basic Statistics

In `datasplitandEDA.py`, output the following logs for the raw training data: `data.head()`, `data.info()`, `data.describe()`, number of missing values, and proportion of missing values. Save the resulting file as `data_describe_summary.csv` in each dataset's directory.

Below is an example from the Low-res-less-variables dataset using `ice_thickness` as the label:

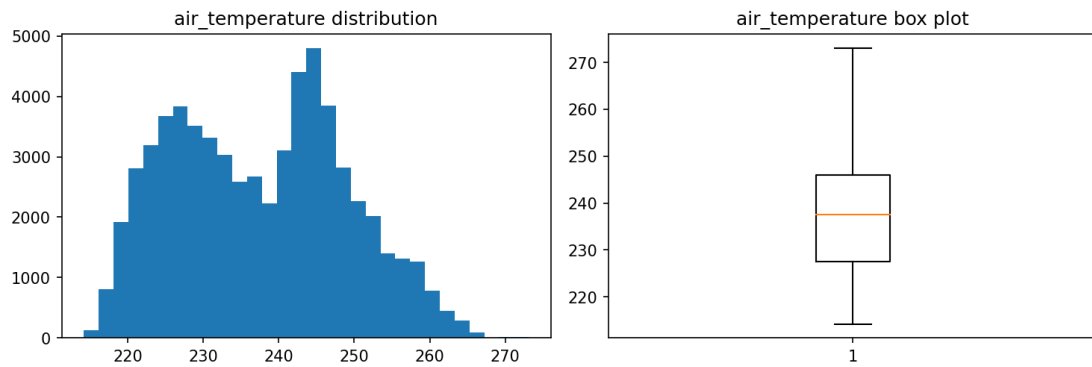
data_describe_summary									
	count	mean	std	min	25%	50%	75%	max	
year	62605.0	2048.356680776300	19.62121062931100	2015.0	2031.0	2048.0	2065.0	2082.0	
x_coordinate	62605.0	451783.6722306530	1172030.3381903600	-2432000.0	-486400.0	608000.0	1459200.0	2675200.0	
y_coordinate	62605.0	77489.52959028830	1003126.9153748500	-2067200.0	-729600.0	121600.0	851200.0	2188800.0	
precipitation	62605.0	208.8470233918	276.77546053600900	-108.359266188	44.2476844788	98.0099639893	251.934860229	2459.45605469	
air_temperature	62605.0	237.54867608379800	11.510238339904000	214.175415039	227.551956177	237.588165283	246.036178589	273.111785889	
ocean_temperature	62605.0	272.325862614285	0.5461632284764640	271.191192627	271.924499512	272.236175537	272.607666016	275.000488281	
upper_left_precipitation	62605.0	164.81362225985300	231.6699438772850	-108.359266188	35.0633239746	78.2778091431	196.929824829	2459.45605469	
upper_left_air_temperature	62605.0	219.36646871074000	61.938116205963500	0.0	225.278747559	234.288879395	244.305114746	273.111785889	
upper_left_ocean_temperature	62605.0	252.7416289978540	70.32136837272450	0.0	271.849975586	272.169891357	272.550689697	274.443939209	
top_precipitation	62605.0	169.29598435156000	225.40547570005300	-108.359266188	36.7137947083	81.8392944336	208.366394043	2294.9050293	
top_air_temperature	62605.0	222.93057239496800	56.12383619100130	0.0	225.724060059	234.980285645	244.729644775	266.438110352	
top_ocean_temperature	62605.0	256.64304964632100	63.44172866501870	0.0	271.86807251	272.191040039	272.580963135	275.000488281	
upper_right_precipitation	62605.0	156.8769297216300	203.94471057527100	-108.359266188	34.6502075195	77.7619018555	196.60144043	1816.53466797	
upper_right_air_temperature	62605.0	218.1699912408870	63.701822615928400	0.0	225.122589111	234.143707275	244.340072832	266.81262207	
upper_right_ocean_temperature	62605.0	251.4692674197980	72.44206897295090	0.0	271.859710693	272.193359375	272.585449219	275.000488281	
left_precipitation	62605.0	180.04837362999000	258.29127877575290	-108.359266188	37.4941062927	83.0801620483	209.141571045	2459.45605469	
left_air_temperature	62605.0	224.49185184710000	53.2228276918421	0.0	225.923843384	235.270935059	244.717330933	273.111785889	
left_ocean_temperature	62605.0	258.3669285176100	60.00750233374530	0.0	271.87008667	272.185577393	272.558532715	274.274291992	
right_precipitation	62605.0	170.78114176306000	218.0682936711750	-108.359266188	37.5914649963	83.5234375	212.064819336	1967.79467773	
right_air_temperature	62605.0	224.53498979648700	53.23638519953510	0.0	225.923843384	235.271621704	244.829086304	267.590484619	
right_ocean_temperature	62605.0	258.39510527197200	60.014204578905900	0.0	271.886352539	272.209533691	272.592834473	275.000488281	
lower_left_precipitation	62605.0	171.89783408031800	255.56585453653500	-108.359266188	34.5181045532	76.7882995605	191.321578979	2342.63427734	
lower_left_air_temperature	62605.0	218.12434242162600	63.69672178732290	0.0	225.122589111	234.052459717	244.096954346	273.111785889	
lower_left_ocean_temperature	62605.0	251.42454401946500	72.42888222266600	0.0	271.849884033	272.160675049	272.522857666	274.274291992	
bottom_precipitation	62605.0	180.94391006889800	261.69733009000000	-65.8212127686	36.7096328735	81.4745483398	203.799255371	2459.45605469	
bottom_air_temperature	62605.0	222.9005712210120	56.12077461682580	0.0	225.724060059	234.980285645	244.536315918	273.111785889	
bottom_ocean_temperature	62605.0	256.62874721493300	63.437981616239500	0.0	271.873260498	272.187225342	272.559509277	275.000488281	
lower_right_precipitation	62605.0	167.83317732762500	231.61369374538600	-108.359266188	35.0568275452	77.9970092773	196.071472168	2335.80810547	
lower_right_air_temperature	62605.0	219.33260947292000	61.918369125347600	0.0	225.278747559	234.288879395	244.258377075	267.590484619	
lower_right_ocean_temperature	62605.0	252.75712894086100	70.32567005052600	0.0	271.868286133	272.18850708	272.569244385	275.000488281	
precipitation_3_years_ago	62605.0	197.78007804839300	272.54389754908800	-108.359266188	39.0327529907	89.2989196777	237.613952637	2459.45605469	
air_temperature_3_years_ago	62605.0	226.54826353664300	50.98969743425210	0.0	226.329376221	236.395858765	245.630111694	273.111785889	
ocean_temperature_3_years_ago	62605.0	259.7934481550490	57.04131552857310	0.0	271.876647949	272.195770264	272.577667236	275.000488281	
precipitation_2_years_ago	62605.0	201.40497763691400	273.7873138900730	-108.359266188	40.6939849854	92.4044342041	242.837615967	2459.45605469	
air_temperature_2_years_ago	62605.0	230.17174722710900	42.582963316095900	0.0	226.73008728	236.788692017	245.772155762	273.111785889	
ocean_temperature_2_years_ago	62605.0	263.9239060814360	47.07431894759470	0.0	271.892456055	272.209106445	272.58706665	275.000488281	
precipitation_1_year_ago	62605.0	205.1605672779940	275.3511394241540	-108.359266188	42.5544586182	95.2460098267	247.490127563	2459.45605469	
air_temperature_1_year_ago	62605.0	233.81507567360000	31.57444256199730	0.0	227.168258667	237.182388306	245.913543701	273.111785889	
ocean_temperature_1_year_ago	62605.0	268.07264596405800	33.75746988720320	0.0	271.90814209	272.223449707	272.597076416	275.000488281	
ice_thickness	62605.0	1918.8392251958700	1071.6295773098800	0.0100353881717	982.284790039	2073.45605469	2837.94873047	4586.17773438	

4.2. Variable Distribution

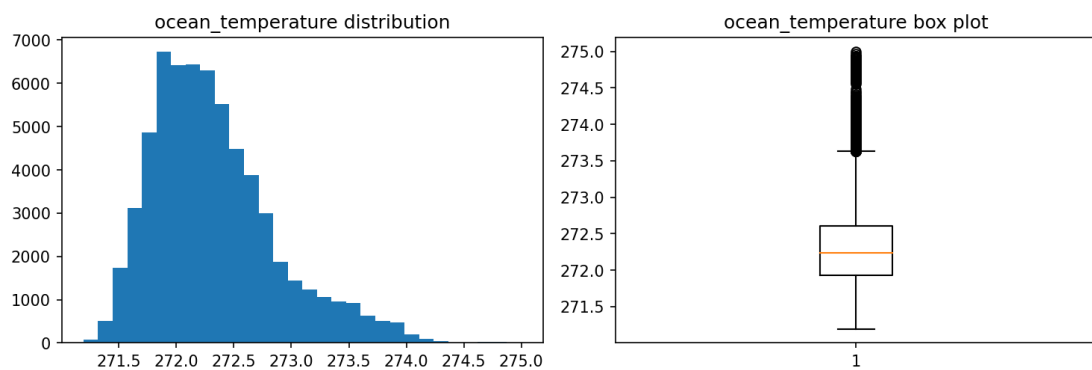
The distribution plot for each variable in the training set is stored in the “distribution_plot” directory of that dataset.

Below are the distribution plots of several variables from the Low-res-less-variables dataset, using ice_thickness as the label:

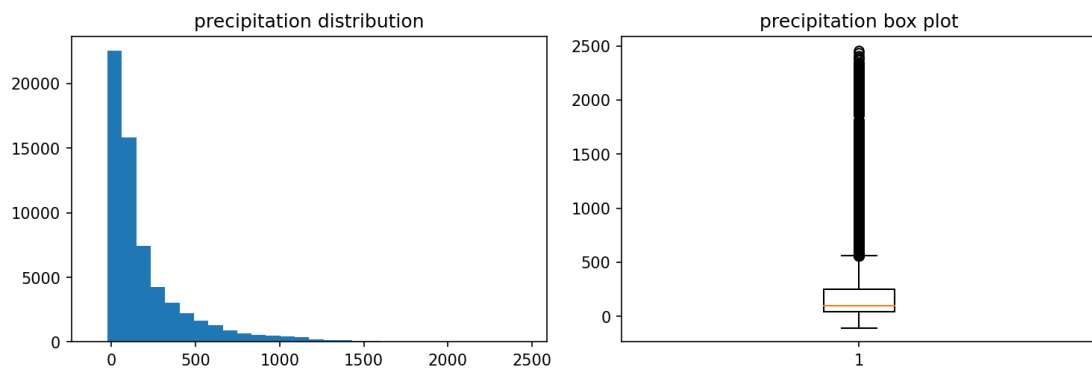
air_temperature



ocean_temperature



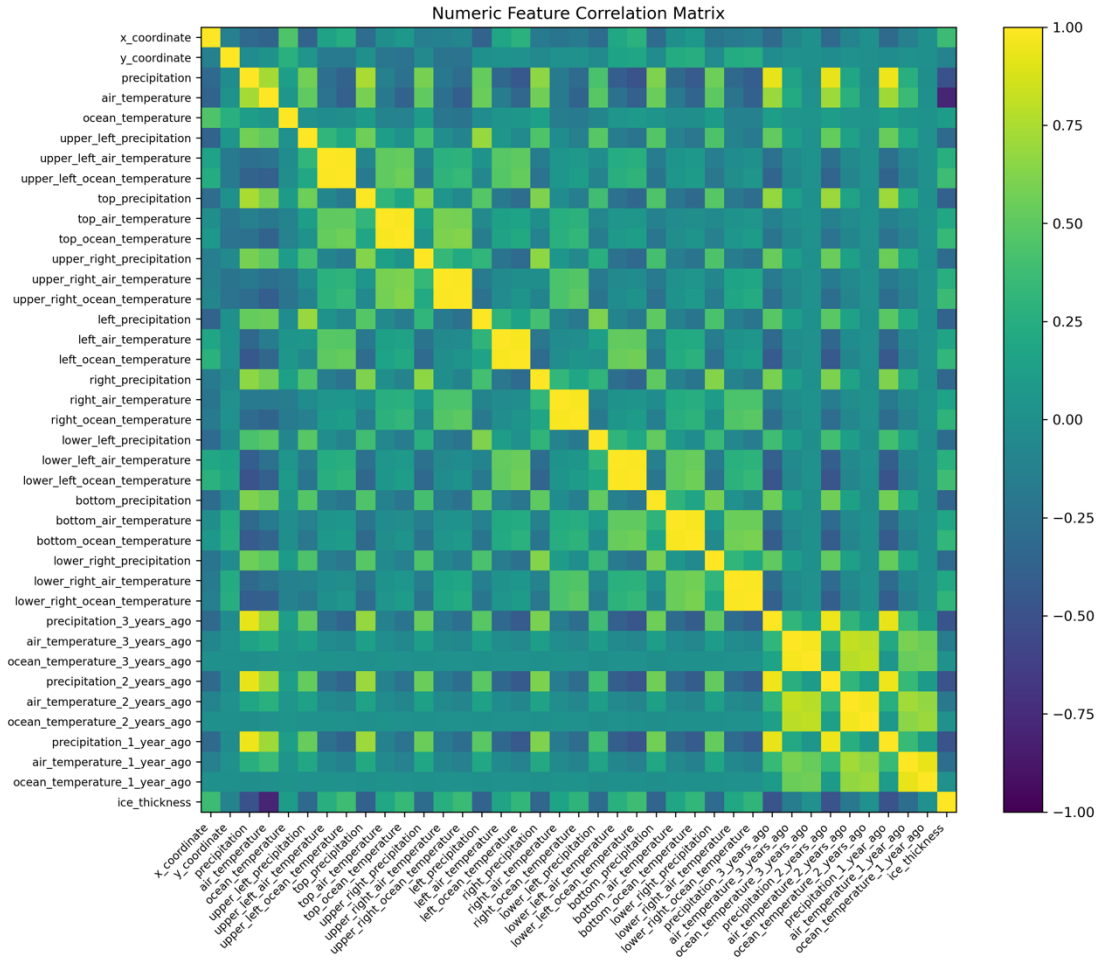
precipitation



4.3. Correlation Heat Map

Compute the Pearson correlation coefficients between features and save the result as `correlation_matrix.png` in the corresponding directory.

Below is an example from the Low-res-less-variables dataset using `ice_thickness` as the label:

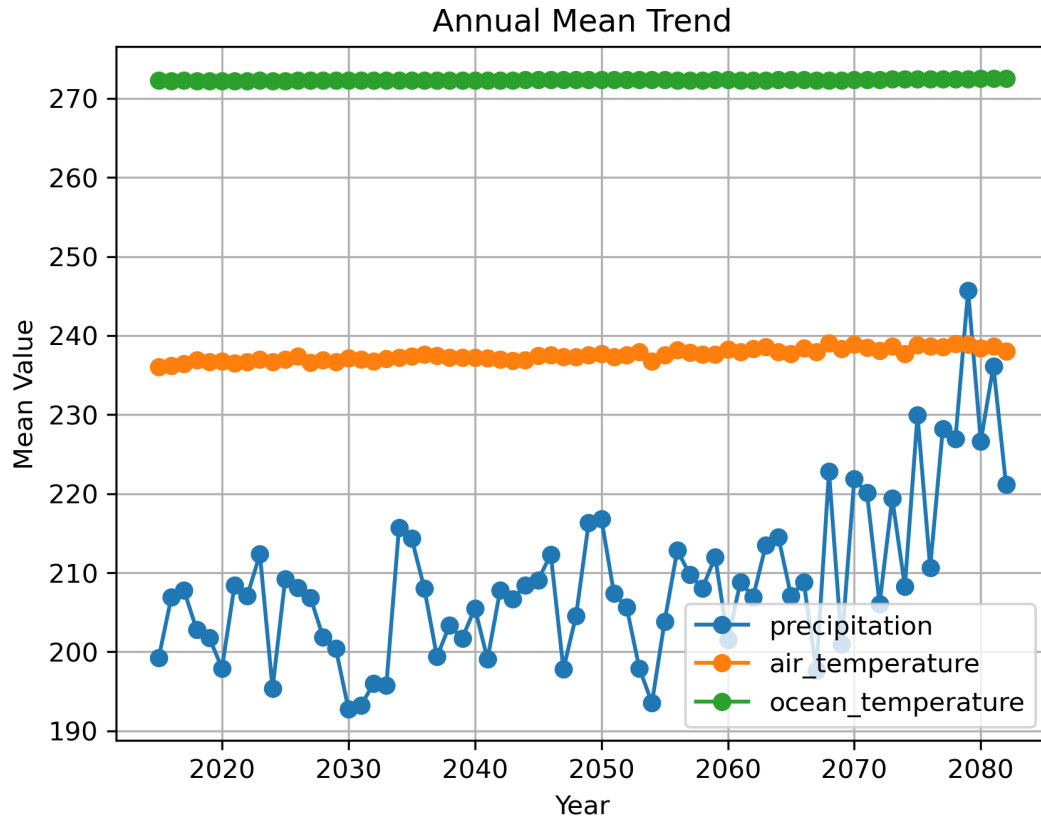


It can be observed that precipitation is highly positively correlated with precipitation from the preceding three years, and exhibits a strong negative correlation with ocean temperature at the eight surrounding locations. In future work, feature optimization or dimensionality reduction may be considered.

4.4. Time Trend

Plot the annual averages of “precipitation,” “air_temperature,” and “ocean_temperature,” and save the figure as “yearly_trend.png” in the corresponding dataset directory.

Below is an example from the Low-res-less-variables dataset using ice_thickness as the label:

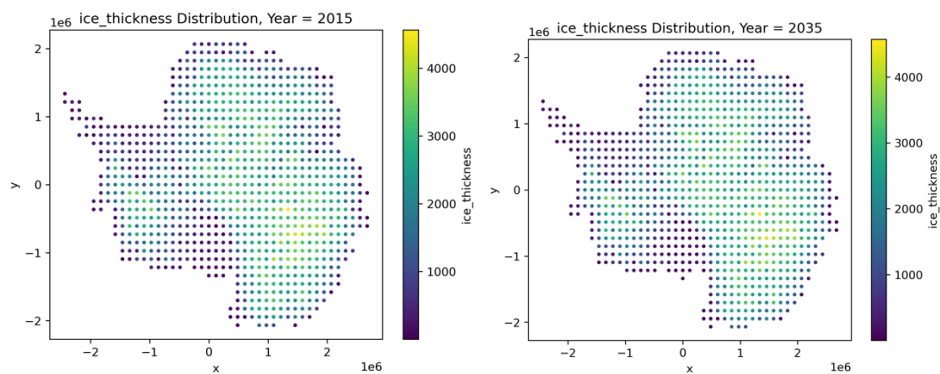


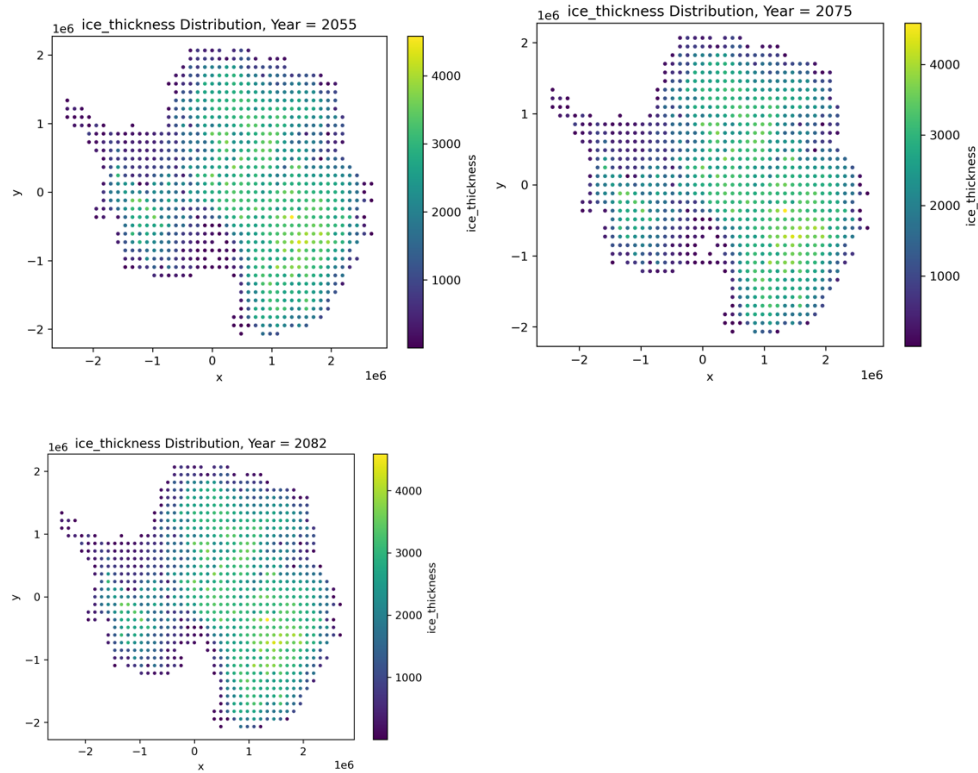
It can be observed that precipitation exhibits almost no change over successive years, whereas air temperature shows a slight upward trend. In contrast, ocean temperature increases with considerable fluctuations. Therefore, temporal information has a measurable impact on the predictions.

4.5. Geographic Distribution

Generate geographic distribution plots for each year and save them in the dataset's geographic_distribution directory.

Below are the geographic distribution plots of several years from the Low-res-less-variables dataset, using ice_thickness as the label:





5. DATA PREPROCESSING

5.1. Outlier Handling

Outliers are clipped based on the interquartile range (IQR). For each numerical column, values are constrained to the interval $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$, removing extreme noise and reducing the impact of outliers on model training.

5.2. Missing Value Imputation

Use scikit-learn's SimpleImputer to batch-impute missing values in the training and test sets, with the default strategy set to "median."

5.3. Standardization

Use scikit-learn's StandardScaler to perform z-score normalization on the numerical columns of the training and test sets.

5.4. Comparison Between Preprocessed Data and Raw Data

Below is an example from the Low-res-less-variables dataset using ice_thickness as the label:

Processed data				Raw data			
	count	mean	std	min	25%	50%	75%
year	62905.0	-3.59943710004566E-15	1.0000079866782300	-1.7000465314000560	-0.8645641729864470	-0.018178471645030700	0.8483778017437
x_coordinate	62905.0	-1.61114653115461E-17	1.0000079866781600	-2.4695222323036200	-0.800486698651970	0.332879963303000	0.8936548603587230
y_coordinate	62905.0	-2.72390793997962E-18	1.0000079866780200	-2.13982712470284100	-0.6048601216021400	0.04397332164987100	0.7713034844806360
precipitation	62905.0	-3.6959649785678E-17	1.0000079866781700	-1.6270971656803200	-0.7551282625550530	-0.4417938544464680	0.4515809844253520
air_temperature	62905.0	-1.257537149855728E-15	1.0000079866782200	-2.030665823903800	-0.6685137051181530	0.00543081641087400	0.7373309856076210
ocean_temperature	62905.0	1.1181235652328E-14	1.0000079866782300	-2.151092403600360	-0.7513867595868710	0.556007218275580	2.508602651440000
upper_left_precipitation	62905.0	-5.89259746795516E-17	1.0000079866782000	-1.790486520068100	-0.7405774782104880	-0.462428627897300	0.4434821656660400
upper_left_air_temperature	62905.0	-1.16038417824332E-15	1.0000079866782000	-2.5037297238719000	-0.560256593059790	0.05961917457467670	0.7357772220713820
upper_left_ocean_temperature	62905.0	-4.612188229817788E-14	1.0000079866782400	-2.2137814440282900	-0.5424271769864840	-0.23478153005881020	0.7489800167852910
top_precipitation	62905.0	1.4164321287394E-16	1.0000079866782800	-1.740871900374100	-0.743729497739680	0.446774747846880	2.252516462087120
top_air_temperature	62905.0	-1.48106591934891E-15	1.0000079866782400	-2.6400798549253900	-0.6098125530359680	0.04690866283182050	0.7489800167852910
top_ocean_temperature	62905.0	1.2845707165918E-13	1.0000079866782300	-2.2956187038689700	-0.5780865165688750	-0.4616252232381220	0.56401632033145210
upper_right_precipitation	62905.0	6.80979844694050E-17	1.0000079866783100	-1.7900140353587300	-0.7403485913561450	-0.42387816760464310	0.4484083051570700
upper_right_air_temperature	62905.0	1.63434476398777E-16	1.0000079866782100	-2.4822125189781870	-0.5438783143704570	0.08278646232053450	0.7483538218004670
upper_right_ocean_temperature	62905.0	7.89570114868759E-14	1.0000079866781800	-2.1917165076148860	-0.5337288075884140	-0.02555682766136580	0.571627037835219
left_precipitation	62905.0	2.1791235519837E-17	1.0000079866782400	-1.744734143482700	-0.743880139408120	-0.4310583576051660	0.4340254546205510
left_air_temperature	62905.0	4.539846566627E-18	1.0000079866782100	-2.862008464581780	-0.6341782972326550	0.4461406500586100	0.737707144334690
left_ocean_temperature	62905.0	-1.7894592270594E-14	1.0000079866782200	-2.3446854302238700	-0.599186463803500	-0.06199751883212520	0.5991867045155840
right_precipitation	62905.0	-5.60180253682251E-17	1.0000079866781800	-1.7320710470427390	-0.7439547799703530	-0.433005009715730	0.437180044005030
right_air_temperature	62905.0	-2.65871821078548E-15	1.0000079866782400	-2.668115745088560	-0.634863325982746	0.04552126661835040	0.740686004619180
right_ocean_temperature	62905.0	-3.5204808025225E-14	1.0000079866782000	-2.43208691242710	-0.5981802656186750	-0.0644398047308850	0.598180513870740
lower_left_precipitation	62905.0	7.808380446024E-17	1.0000079866781400	-1.797327480722380	-0.7384483978791000	-0.4235878760430000	0.4278334215276810
lower_left_air_temperature	62905.0	-1.76146046785348E-16	1.0000079866782100	-2.4639111625472770	-0.544318248242020	0.05785674400968300	0.735408772847344
lower_left_ocean_temperature	62905.0	1.5625643380505E-14	1.000007986678280	-2.20153097876556	-0.5311580673430390	-0.1739711401147690	0.5818118521777550
bottom_precipitation	62905.0	-1.0550607178228E-16	1.0000079866782100	-1.453306236221780	-0.741020498687400	-0.4207863598006650	0.420796023050040
bottom_air_temperature	62905.0	2.8651646228187E-15	1.0000079866782400	-2.624456680378590	-0.6102517188871500	0.0504431060451720	0.732537590770180
bottom_ocean_temperature	62905.0	5.8146534825418E-14	1.0000079866782000	-2.309819565100850	-0.579728863711980	-0.06520796651310870	0.573070194465620
lower_right_precipitation	62905.0	-1.04416471032552E-16	1.0000079866782100	-1.770138028464350	-0.7396297178828380	-0.42758765714600	0.4230654548248740
lower_right_air_temperature	62905.0	1.86153708231941E-15	1.0000079866781800	-2.405839001807180	-0.5691585675861620	0.0657865537374110	0.573731523208160
lower_right_ocean_temperature	62905.0	-1.0548128714835E-13	1.000007986678150	-2.224817848282850	-0.543138414038370	-0.0209751405248877	0.57979562265900
precipitation_3_years_ago	62905.0	-1.32688652787554E-17	1.0000079866782000	-1.816987028129180	-0.745208858344170	-0.4473517770130100	0.4524068071912270
air_temperature_3_years_ago	62905.0	2.7529269500607E-15	1.0000079866782200	-2.739297178827850	-0.650654859575430	0.0645998073770880	0.713882240409280
ocean_temperature_3_years_ago	62905.0	1.3594116599128E-14	1.0000079866782100	-2.387184479501700	-0.6234418275985670	0.02665045953448100	0.789388759358720
precipitation_2_years_ago	62905.0	2.2517938704082E-16	1.0000079866781900	-1.6201142491577790	-0.7448118720954170	-0.446550474098600	0.554089890718810
air_temperature_2_years_ago	62905.0	1.47041028738989E-15	1.0000079866782100	-2.867028709885070	-0.6234418275985670	0.02665045953448100	0.789388759358720
ocean_temperature_2_years_ago	62905.0	-7.3828080850507E-15	1.0000079866782200	-2.47186079174100	-0.6504475282641570	0.027013219892390	0.559385941912270
precipitation_1_year_ago	62905.0	-8.698992706000E-17	1.0000079866782200	-1.623600824874900	-0.7514523281800210	-0.4449427687801380	0.423883419055490
air_temperature_1_year_ago	62905.0	-5.7024421190233E-16	1.0000079866782300	-3.061885783897330	-0.7888447257398830	0.0211007151121440	0.7270135973288190
ocean_temperature_1_year_ago	62905.0	2.88153114277311E-14	1.0000079866782000	-2.863813912712610	-0.729064027086660	-0.1257346267312200	0.556215821832240
ice_thickness	62905.0	-1.191770872374108E-16	1.0000079866782200	-1.789586523285180	-0.6739864943219840	0.1442831251041950	0.6578814837519180
				Row data			
	count	mean	std	min	25%	50%	75%
year	62905.0	-3.59943710004566E-15	1.0000079866782300	-1.7000465314000560	-0.8645641729864470	-0.018178471645030700	0.8483778017437
x_coordinate	62905.0	-1.61114653115461E-17	1.0000079866781600	-2.4695222323036200	-0.800486698651970	0.332879963303000	0.8936548603587230
y_coordinate	62905.0	-2.72390793997962E-18	1.0000079866780200	-2.13982712470284100	-0.6048601216021400	0.04397332164987100	0.7713034844806360
precipitation	62905.0	-3.6959649785678E-17	1.0000079866781700	-1.6270971656803200	-0.7551282625550530	-0.4417938544464680	0.4515809844253520
air_temperature	62905.0	-1.257537149855728E-15	1.0000079866782200	-2.030665823903800	-0.6685137051181530	0.00543081641087400	0.7373309856076210
ocean_temperature	62905.0	1.1181235652328E-14	1.0000079866782300	-2.151092403600360	-0.7513867595868710	0.556007218275580	2.508602651440000
upper_left_precipitation	62905.0	-5.89259746795516E-17	1.0000079866782000	-1.790486520068100	-0.7405774782104880	-0.462428627897300	0.4434821656660400
upper_left_air_temperature	62905.0	-1.16038417824332E-15	1.0000079866782000	-2.5037297238719000	-0.560256593059790	0.05961917457467670	0.7357772220713820
upper_left_ocean_temperature	62905.0	-4.612188229817788E-14	1.0000079866782400	-2.2137814440282900	-0.5424271769864840	-0.23478153005881020	0.7489800167852910
top_precipitation	62905.0	1.4164321287394E-16	1.0000079866782800	-1.740871900374100	-0.743729497739680	0.446774747846880	2.252516462087120
top_air_temperature	62905.0	-1.48106591934891E-15	1.0000079866782400	-2.6400798549253900	-0.6098125530359680	0.04690866283182050	0.7489800167852910
top_ocean_temperature	62905.0	1.2845707165918E-13	1.0000079866782300	-2.2956187038689700	-0.5780865165688750	-0.4616252232381220	0.56401632033145210
upper_right_precipitation	62905.0	6.80979844694050E-17	1.0000079866783100	-1.7900140353587300	-0.7403485913561450	-0.42387816760464310	0.4484083051570700
upper_right_air_temperature	62905.0	1.63434476398777E-16	1.0000079866782100	-2.4822125189781870	-0.5438783143704570	0.08278646232053450	0.7483538218004670
upper_right_ocean_temperature	62905.0	7.89570114868759E-14	1.0000079866781800	-2.1917165076148860	-0.5337288075884140	-0.02555682766136580	0.571627037835219
left_precipitation	62905.0	2.1791235519837E-17	1.0000079866782400	-1.744734143482700	-0.743880139408120	-0.4310583576051660	0.4340254546205510
left_air_temperature	62905.0	4.539846566627E-18	1.0000079866782100	-2.862008464581780	-0.6341782972326550	0.4461406500586100	0.737707144334690
left_ocean_temperature	62905.0	-1.7894592270594E-14	1.0000079866782200	-2.3446854302238700	-0.599186463803500	-0.06199751883212520	0.5991867045155840
right_precipitation	62905.0	-5.60180253682251E-17	1.0000079866781800	-1.7320710470427390	-0.7439547799703530	-0.433005009715730	0.437180044005030
right_air_temperature	62905.0	-2.65871821078548E-15	1.0000079866782400	-2.668115745088560	-0.634863325982746	0.04552126661835040	0.740686004619180
right_ocean_temperature	62905.0	-3.5204808025225E-14	1.0000079866782000	-2.43208691242710	-0.5981802656186750	-0.0644398047308850	0.598180513870740
lower_left_precipitation	62905.0	7.808380446024E-17	1.0000079866781400	-1.797327480722380	-0.7384483978791000	-0.4235878760430000	0.4278334215276810
lower_left_air_temperature	62905.0	-1.76146046785348E-16	1.0000079866782100	-2.4639111625472770	-0.544318248242020	0.05785674400968300	0.735408772847344
lower_left_ocean_temperature	62905.0	1.5625643380505E-14	1.000007986678280	-2.20153097876556	-0.5311580673430390	-0.1739711401147690	0.5818118521777550
bottom_precipitation	62905.0	-1.0550607178228E-16	1.0000079866782100	-1.453306236221780	-0.741020498687400	-0.4207863598006650	0.420796023050040
bottom_air_temperature	62905.0	2.8651646228187E-15	1.0000079866782400	-2.624456680378590	-0.6102517188871500	0.0504431060451720	0.732537590770180
bottom_ocean_temperature	62905.0	5.8146534825418E-14	1.0000079866782000	-2.309819565100850	-0.579728863711980	-0.06520796651310870	0.573070194465620
lower_right_precipitation	62905.0	-1.04416471032552E-16	1.0000079866782100	-1.770138028464350	-0.7396297178828380	-0.42758765714600	0.4230654548248740
lower_right_air_temperature	62905.0	1.86153708231941E-15	1.0000079866781800	-2.405839001807180	-0.5691585675861620	0.0657865537374110	0.573731523208160
lower_right_ocean_temperature	62905.0	-1.0548128714835E-13	1.000007986678150	-2.224817848282850	-0.543138414038370	-0.0209751405248877	0.57979562265900
precipitation_3_years_ago	62905.0	-1.32688652787554E-17	1.0000079866782000	-1.816987028129180	-0.745208858344170	-0.4473517770130100	0.4524068071912270
air_temperature_3_years_ago	62905.0	2.7529269500607E-15	1.0000079866782200	-2.739297178827850	-0.650654859575430	0.0645998073770880	0.713882240409280
ocean_temperature_3_years_ago	62905.0	1.3594116599128E-14	1.0000079866782100	-2.387184479501700	-0.6234418275985670	0.02665045953448100	0.789388759358720
precipitation_2_years_ago	62905.0	2.2517938704082E-16	1.0000079866781900	-1.6201142491577790	-0.7448118720954170	-0.446550474098600	0.55408

Raw Data

```

=====Number of missing values=====
year                                0
x_coordinate                        0
y_coordinate                        0
precipitation                      0
air_temperature                    0
ocean_temperature                  0
upper_left_precipitation           0
upper_left_air_temperature         0
upper_left_ocean_temperature       0
top_precipitation                  0
top_air_temperature                0
top_ocean_temperature              0
upper_right_precipitation          0
upper_right_air_temperature        0
upper_right_ocean_temperature      0
left_precipitation                 0
left_air_temperature               0
left_ocean_temperature             0
right_precipitation                0
right_air_temperature              0
right_ocean_temperature            0
lower_left_precipitation           0
lower_left_air_temperature         0
lower_left_ocean_temperature       0
bottom_precipitation               0
bottom_air_temperature             0
bottom_ocean_temperature           0
lower_right_precipitation          0
lower_right_air_temperature        0
lower_right_ocean_temperature      0
precipitation_3_years_ago          0
air_temperature_3_years_ago        0
ocean_temperature_3_years_ago      0
precipitation_2_years_ago          0
air_temperature_2_years_ago        0
ocean_temperature_2_years_ago      0
precipitation_1_year_ago           0
air_temperature_1_year_ago         0
ocean_temperature_1_year_ago       0
ice_thickness                      0
dtype: int64

```

PreProcessed Data

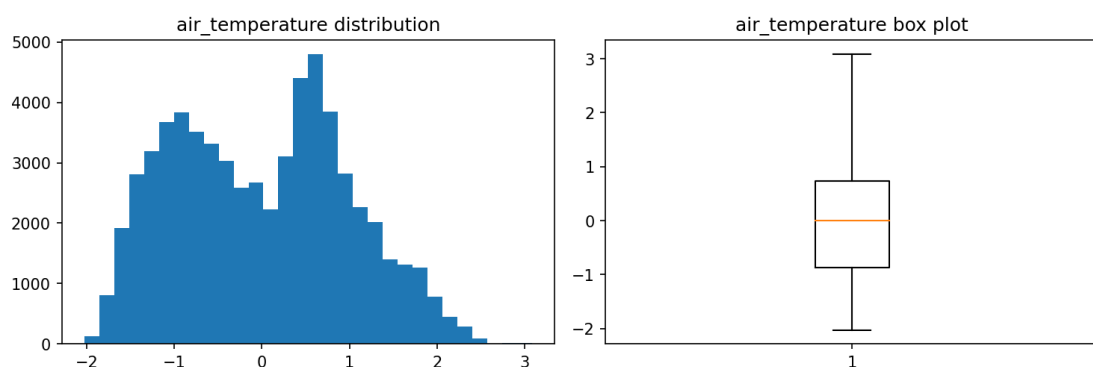
```

=====Number of missing values=====
year                                0
x_coordinate                        0
y_coordinate                        0
precipitation                      0
air_temperature                    0
ocean_temperature                  0
upper_left_precipitation           0
upper_left_air_temperature         0
upper_left_ocean_temperature       0
top_precipitation                  0
top_air_temperature                0
top_ocean_temperature              0
upper_right_precipitation          0
upper_right_air_temperature        0
upper_right_ocean_temperature      0
left_precipitation                 0
left_air_temperature               0
left_ocean_temperature             0
right_precipitation                0
right_air_temperature              0
right_ocean_temperature            0
lower_left_precipitation           0
lower_left_air_temperature         0
lower_left_ocean_temperature       0
bottom_precipitation               0
bottom_air_temperature             0
bottom_ocean_temperature           0
lower_right_precipitation          0
lower_right_air_temperature        0
lower_right_ocean_temperature      0
precipitation_3_years_ago          0
air_temperature_3_years_ago        0
ocean_temperature_3_years_ago      0
precipitation_2_years_ago          0
air_temperature_2_years_ago        0
ocean_temperature_2_years_ago      0
precipitation_1_year_ago           0
air_temperature_1_year_ago         0
ocean_temperature_1_year_ago       0
ice_thickness                      0
dtype: int64

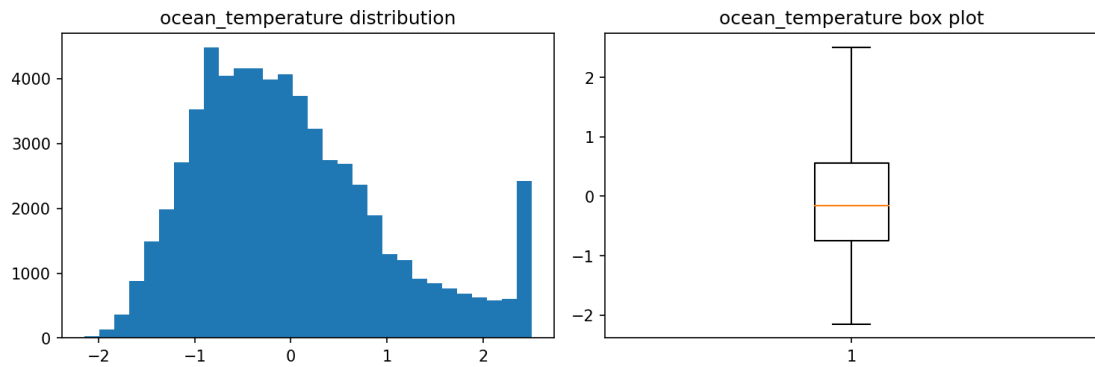
```

Below is some examples distribution plot of the preprocessed data from the Low-res-less-variables dataset, using ice thickness as the label:

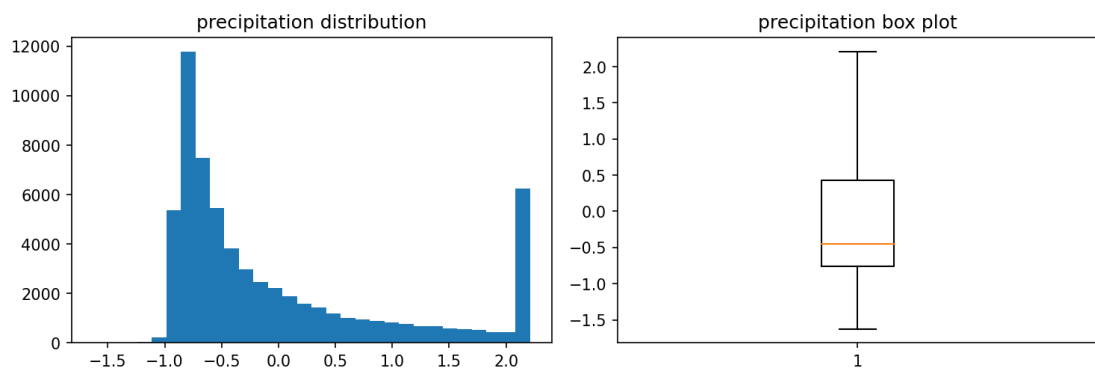
air_temperature



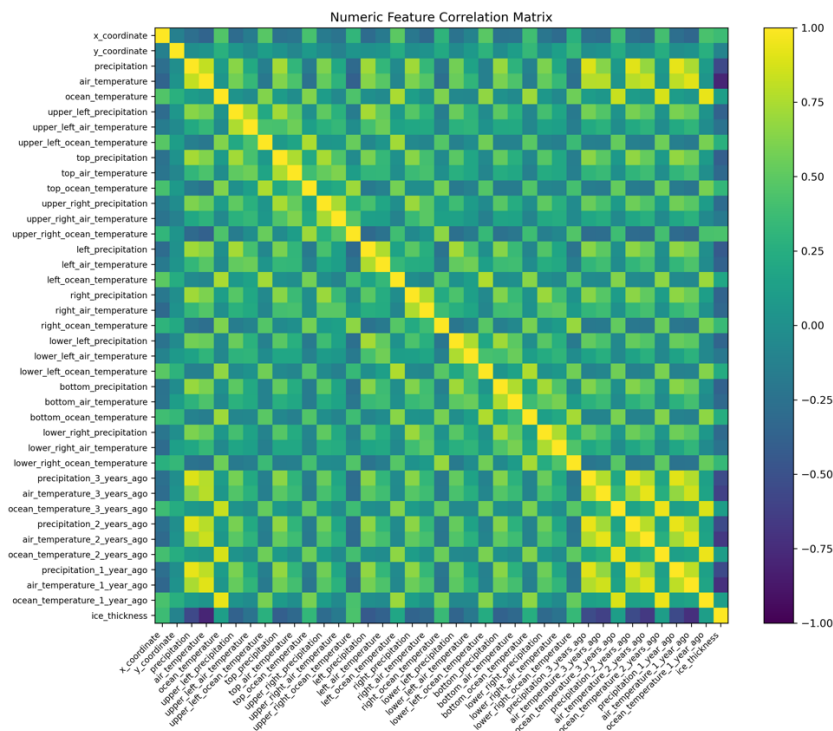
ocean_temperature



precipitation



Below is an example correlation heat map of the preprocessed data from the Low-res-less-variables dataset, using ice thickness as the label:



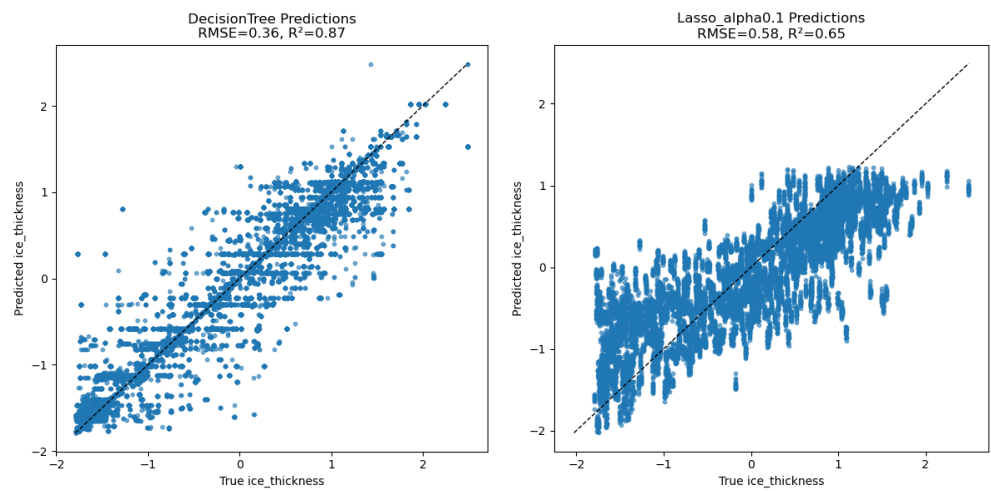
6. MODEL TRAINING AND EVALUATION

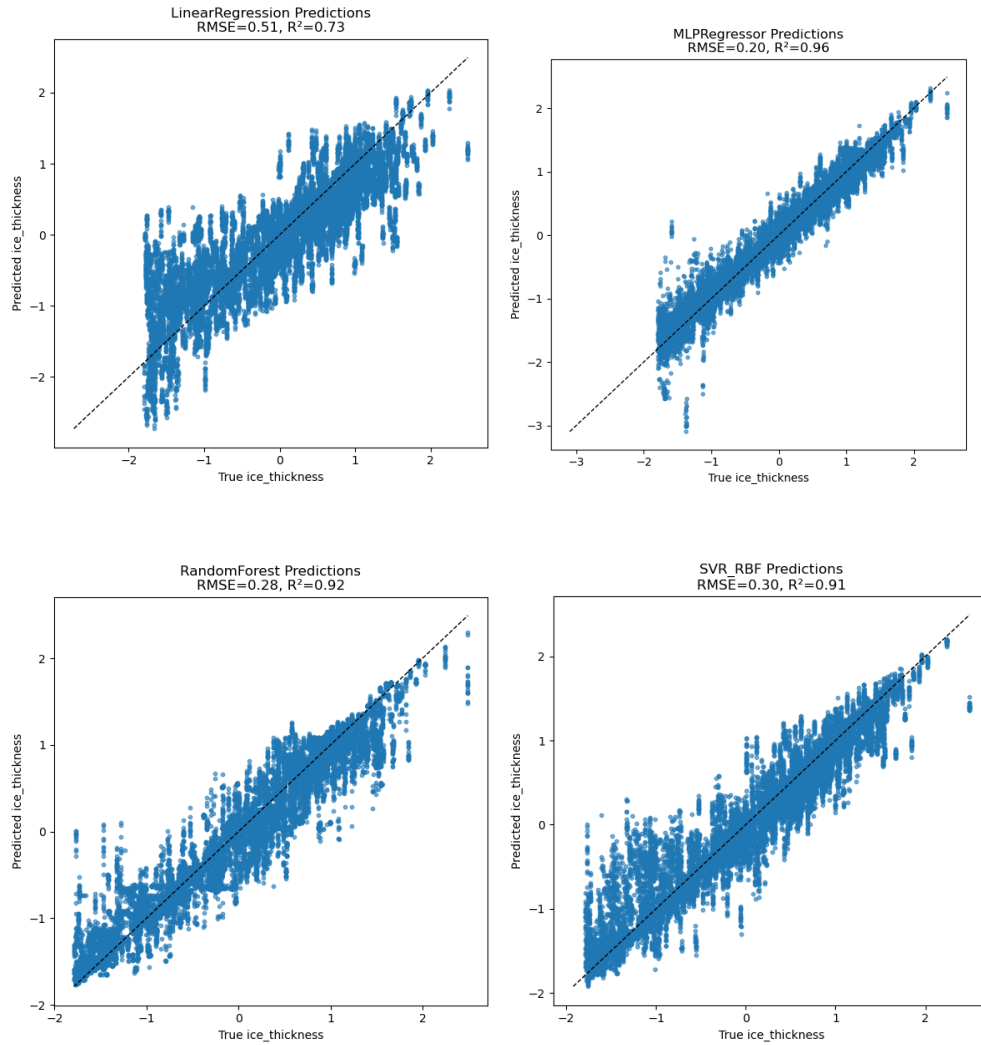
Used the following six regression models to train on the previously engineered feature set, generate predictions, and evaluate their performance:

Model Name	Python Class	Key Hyperparameters
LinearRegression	sklearn.linear_model.LinearRegression	(default settings, no regularization)
Lasso_alpha0.1	sklearn.linear_model.Lasso	alpha=0.1, max_iter=10000
SVR_RBF	sklearn.svm.SVR	kernel='rbf', C=1.0, epsilon=0.1
DecisionTree	sklearn.tree.DecisionTreeRegressor	max_depth=10, random_state=<seed>
RandomForest	sklearn.ensemble.RandomForestRegressor	n_estimators=100, max_depth=10, random_state=<seed>, n_jobs=-1
MLPRegressor	sklearn.neural_network.MLPRegressor	hidden_layer_sizes=(100,50), activation='relu', solver='adam', max_iter=500, random_state=<seed>

A fixed random seed of 42 was used; the model was trained on the entire training set and evaluated on the entire test set. The evaluation metrics used are Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). Lower RMSE values denote better performance, and R^2 values closer to 1 indicate a superior fit.

Below are the prediction results and performance metrics of each model on the Low-Res Less-Variables dataset, using ice_thickness as the target label:





Model Name	RMSE	R^2
LinearRegression	0.510	0.733
Lasso_alpha0.1	0.584	0.650
SVR_RBF	0.298	0.909
DecisionTree	0.359	0.867
RandomForest	0.283	0.918
MLPRegressor	0.200	0.959

The figure above compares each model's performance in terms of RMSE and R^2 . Overall, nonlinear models (SVR, Decision Tree, Random Forest, MLP) clearly outperform linear models (LinearRegression, Lasso). Moreover, MLPRegressor achieves the best results for both RMSE and R^2 , with Random Forest a close second. This indicates that the data contains inherent nonlinear relationships, which is why nonlinear models generally outperform linear ones.

7. USING GENETIC ALGORITHM (GA) TO OPTIMIZE FEATURE SELECTION

According to the previous correlation heat map, many features exhibit strong correlations, which can introduce redundancy during training. Used a GA to optimize feature selection.

The GA is designed as follows:

- **Representation:**

Construct a binary vector of length n (a 0/1 array) to indicate whether each feature is selected, where n is the total number of candidate features. Specifically:

- Each chromosome (individual) is a binary array of the form $[b_0, b_1, \dots, b_{n-1}]$.
- If the i -th bit $b_i = 1$, it means feature i is selected; if $b_i = 0$, that feature is skipped.

For example, if there are 12 candidate features in total, one possible chromosome could be:

$[1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 1]$

This indicates that features 0, 2, 4, 5, 8, and 11 (6 out of 12) are selected.

- **Fitness Function:**

The fitness function is the root mean squared error (RMSE) on the validation set. For each chromosome (feature subset), the corresponding columns are extracted from X_{train} and used to train an MLPRegressor. The model then makes predictions on X_{val} and the RMSE of those predictions is computed. The GA's objective is to find the chromosome that minimizes this validation RMSE.

- **Crossover, Mutation, and Selection Operators:**

Due to the limited computing power of my laptop, the population size is set to $\text{pop_size} = 20$ and the number of generations is set to $N_{\text{GEN}} = 10$.

- **Initialization (Sampling):**

Use `BinaryRandomSampling()` to generate a random binary population matrix of shape $(\text{pop_size} \times n_{\text{features}})$.

- **Crossover:**

Apply the two-point crossover operator `TwoPointCrossover(prob=CROSSOVER_PROB)`. `prob` is set to 0.8 indicates that each pair of parents has an 80% chance of undergoing two-point crossover; otherwise, the parents are copied unchanged.

- **Mutation:**

Apply the bit-flip mutation operator `BitflipMutation(prob=MUTATION_PROB)`. `prob` is set to 0.1 indicates that each gene (bit) has a 10% chance of being flipped ($0 \rightarrow 1$ or $1 \rightarrow 0$).

- **Selection and Replacement:**

Use the default tournament selection combined with elitism to form the next generation.

The GA optimization is implemented in the `GAFeatureSelection.py` script. The previously generated `preprocessed_train` and `preprocessed_test` datasets are loaded. The `preprocessed_train` data is split into training and validation sets according to the `VALID_SIZE` ratio for GA training. The model and parameters used are the same as in earlier experiments: `MLPRegressor(hidden_layer_sizes=(100,50))`,

activation='relu', solver='adam', max_iter=500, early_stopping=True, random_state=RANDOM_SEED). Once training is complete, the selected feature subset is printed. That subset is then applied to both the full training set and the full test set. Finally, an MLPRegressor is trained on the entire training set (using only the chosen features) and evaluated on the entire test set.

Below are the results on the Low-Res Less-Variables dataset using ice_thickness as the label:

```

=====
n_gen | n_eval | f_avg | f_min
=====
1 | 20 | 0.2110059691 | 0.1685548639
2 | 40 | 0.1872074507 | 0.1682090155
3 | 60 | 0.1770580136 | 0.1664515768
4 | 80 | 0.1698861026 | 0.1405505990
5 | 100 | 0.1625795227 | 0.1405505990
6 | 120 | 0.1576788447 | 0.1405505990
7 | 140 | 0.1529156541 | 0.1405505990
8 | 160 | 0.1486323252 | 0.1405505990
9 | 180 | 0.1447786407 | 0.1403481757
10 | 200 | 0.1413997142 | 0.1319374789

=== GA Feature Selection Results ===
Best RMSE (validation set): 0.1319
Number of selected features: 23 / 38
List of selected features:
- x_coordinate
- y_coordinate
- precipitation
- air_temperature
- ocean_temperature
- upper_left_air_temperature
- top_air_temperature
- top_ocean_temperature
- upper_right_air_temperature
- left_precipitation
- left_air_temperature
- right_precipitation
- lower_left_precipitation
- lower_left_air_temperature
- lower_left_ocean_temperature
- bottom_precipitation
- bottom_air_temperature
- bottom_ocean_temperature
- lower_right_air_temperature
- lower_right_ocean_temperature
- precipitation_3_years_ago
- precipitation_2_years_ago
- ocean_temperature_1_year_ago

```

Test Set RMSE: 0.1981

Test Set r2: 0.9597

Based on the experimental results, the new feature combination yields improved performance on both RMSE and R^2 metrics (RMSE is now 0.1981 versus 0.2000 previously, and R^2 is now 0.9597 versus 0.9590 previously). However, because both pop_size and N_GEN were set relatively low, the results indicate that the model has not yet converged. These findings serve as a proof-of-concept implementation of one possible GA-based approach.

- Future Work:

Parameter Tuning: Key GA hyperparameters (pop_size, N_GEN, MUTATION_PROB, and CROSSOVER_PROB) can be further optimized. Adjusting these values (for example, increasing population size and N_GEN) should improve convergence and overall performance.

Model and its hyperparameter tuning: Further refinement of the model architecture and optimization of the hyperparameters can be done to obtain better prediction performance.

Initialization Enhancement: The current initial population is generated randomly via BinaryRandomSampling(). It would be beneficial to incorporate domain knowledge by preselecting certain correlated features into the initial population. By seeding GA with these informed individuals, the search process may converge more quickly and discover higher-quality feature subsets.

Shuffling the Training Set: Since the train/test split was performed chronologically by year, it would be advantageous to randomize the training data before fitting the model in order to eliminate any ordering bias. Although MLPRegressor enables shuffling by default at each epoch, other algorithms may require explicit configuration.